PETROS

Official Desktop Installation Guide & Deployment Manual

VERSION 4.0.0

1. Introduction

Petros is an autonomous AI-driven software engineering and ecosystem operating system. This document provides complete instructions for installing, configuring, and verifying Petros on Microsoft Windows 10 and Windows 11 (64-bit).

2. Hardware & Software Requirements

Component	Minimum Requirement	Recommended Specification
Operating System	Windows 10 (64-bit, Version 1909+)	Windows 11 (64-bit, Latest Build)
Processor (CPU)	Intel Core i5 or AMD Ryzen 5 (4 Cores)	Intel Core i7/i9 or AMD Ryzen 7/9 (8+ Cores)
System Memory (RAM)	8 GB RAM	16 GB to 32 GB RAM (for local LLM inference)
Storage Space	500 MB free space	5 GB SSD storage for local models and workspaces
Local AI Engine	Optional (Cloud fallback available)	Ollama with `qwen3:1.7b` model

3. Windows Installation Methods

Option A: Windows Installer (Petros-Setup.exe) — Recommended

1. Navigate to the `Windows/x64/` directory in the release package.
2. Double-click `Petros-Setup.exe` to start the Inno Setup wizard.
3. Choose your destination directory (default: `%LOCALAPPDATA%\Programs\Petros`).
4. Check **Create a desktop shortcut** if desired.
5. Click **Install**. Setup completes in approximately 3 to 5 seconds.
6. Launch **Petros** from your Windows Start Menu or Desktop shortcut.

> **Zero-Dependency Guarantee:** The installer bundles the complete, official Node.js v24 runtime and standalone server dependencies. You do NOT need to install Node.js, npm, Python, or Git

> separately.

Option B: Portable Edition (Petros-Portable.zip)

1. Extract `Petros-Portable.zip` to any local or removable drive (e.g., `C:\Tools\Petros`).
2. Double-click `Petros.exe` inside the extracted folder.
3. Petros initializes in standalone mode without modifying Windows system registry keys.

4. System Architecture & Port Mapping

Petros automatically orchestrates three local network services upon launch:

- **Port 3000 (Core Server):** Express application server serving the React 19 interface, REST APIs, and App Factory.
- **Port 20131 (Laya Router Gateway):** OpenAI-compatible local AI gateway that routes inference through local models or cascades.
- **Port 11434 (Ollama Inference):** Automatically detected if Ollama is running locally for zero-cost offline AI intelligence.

5. Data Storage & User Privacy

Petros strictly isolates application binaries from user projects and private encryption keys:

Data Category	Windows Filesystem Path	Preserved on Uninstall?
Application Binaries	`%LOCALAPPDATA%\Programs\Petros`	Cleanly removed
User Wallets & Encryption Keys	`%APPDATA%\Petros`	**Yes (100% Preserved)**
Generated Projects & Workspace	`%USERPROFILE%\Desktop\petros`	**Yes (100% Preserved)**

6. Safe Mode & Troubleshooting

- **Port 3000 Conflict:** If port 3000 is occupied by another application, close the conflicting process or launch `scripts\run-petros-windows.bat`.
- **Safe Mode Console:** Launch *Petros (Safe Mode Console)* from the Start Menu to view real-time HTTP requests and AI agent logs.
- **Ollama Offline:** Download Ollama from https://ollama.com and run `ollama pull qwen3:1.7b`.

7. Uninstallation

To uninstall, navigate to **Windows Settings > Apps > Installed apps**, select **Petros Autonomous Engineering & Operations AI**, and click **Uninstall**. All application files and shortcuts are removed while your workspace and wallets remain untouched.

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